SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA5116
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage: **40%**

Total Marks: 50

QUESTIONS:

Annexure A

Analytical Problem Solving

Code of Conduct:

I Saaika K S [1925210397] *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

a. Compute n ϕ(n)\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i. Calculate total encryption time for each algorithm
ii. Analyze trade-offs between security and performance
iii. Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison
Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Analytical Problem solving

1) RSA Encryption Analysis

Given:-

$p = 11$

$q = 13$

Public exponent $e = 7$

Message $M = 9$

Problem Understanding:-

RSA is an asymmetric cryptographic algorithm that uses a public key for encryption and a private key for decryption. The objective is to calculate n Euler's totient function $\phi(n)$ the private key d encrypt the given message, decrypt the ciphertext, and analyze the effect of changing e.

Method selection:-

The following RSA formula

$$n = p \times q$$

$$\phi(n) = (p-1)(q-1)$$

The private key d satisfies:

$$e \times d \equiv 1 \pmod{\phi(n)}$$

Encryption:

$$C = M^e \bmod n$$

Decryption:

$$M = C^d \bmod n$$

Solution process:

Step1: calculate n

$$n = p \times q = 11 \times 13$$

$$\boxed{n = 143}$$

Step 2: calculate Euler's totient function

$$\phi(n) = (p-1)(q-1)$$

$$= (11-1)(13-1)$$

$$= 10 \times 12$$

$$\phi(n) = 120$$

Step 3: Find the private key d.

Given $e = 7$ find d such that

$$7d \equiv 1 \pmod{120}$$

$$d = 103$$

$$= 7 \times 103$$

$$= 721$$

$$721 = 6 \times 120 + 1$$

$$\boxed{d = 103}$$

The public key is $(e, n) = (7, 143)$ and the private key is $(d, n) = (103, 143)$

Step 4: Encrypt the message $M = 9$

$$C = 9^7 \bmod 143$$

using modular arithmetic:

$$9^2 = 81$$

$$9^4 \bmod 143 = 81^2 \bmod 143 = 126$$

$$9^7 \bmod 143 = (9^4 \times 9^2 \times 9) \bmod 143$$

$$= (126 \times 81 \times 9) \bmod 143$$

$$\boxed{C = 48}$$

Step 5: Decrypt the ciphertext

$$M = 48^{103} \bmod 143$$

$$\boxed{M = 9}$$

Interpretation of Results:-

RSA successfully encrypts the original message 9 into ciphertext 48. using the private key exponent 103 the ciphertext is decrypted back to 9. This confirms the correctness of RSA process. In practical applications, large key sizes and secure Implementations are necessary to protect Information.

Conclusion:

RSA provides confidentiality through public-key encryption and private-key decryption. The calculations demonstrate the working of RSA and the importance of choosing appropriate key parameters.

4) Diffie-Hellman key Exchange Analysis:-

Given:

prime number $p = 23$

Generator $g = 5$

User A's private key $a = 6$

User B's private key $b = 15$

Problem Understanding:-

Diffie-Hellman is a key exchange algorithm that allows two users to establish a common secret key over a public communication channel. The objective is to calculate both public keys, determine the shared secret keys, explain a Man-in the middle (MITM) attack, and suggest a method to prevent the attack.

Method selection:-

The public key formula are:

For user A:

$$A = g^a \bmod p$$

For user B:

$$B = g^b \bmod p$$

The shared secret key is calculated using either public key:-

$$K = B^a \bmod p \text{ (or)}$$

$$K = A^b \bmod p$$

Solution process:-

Step1: calculate user A's public key

Given $g = 5$ $a = 6$ and $p = 23$

$$A = 5^6 \bmod 23$$

$$5^2 = 25 \bmod 23 = 2$$

$$5^4 \bmod 23 = 2^2 \bmod 23 = 4$$

$$5^6 \bmod 23 = (5^4 \times 5^2) \bmod 23$$

$$= (4 \times 2) \bmod 23$$

$$\boxed{A = 8}$$

Step2: calculate user B's public key

$$B = 5^{15} \bmod 23$$

using modular exponentiations:-

$$5^2 \bmod 23 = 2$$

$$5^4 \bmod 23 = 4$$

$$5^8 \bmod 23 = 16$$

Since $15 = 8 + 4 + 2 + 1$

$$5^{15} \bmod 23 = (16 \times 4 \times 2 \times 5) \bmod 23$$

$$= (640) \bmod 23$$

$$\boxed{B = 19}$$

step 3: calculate the shared secret key:-

user A calculates the secret using user B's public key.

$$K = B^a \bmod p$$
$$K = 19^6 \bmod 23$$

Since
$19^2 \bmod 23 = 16$
$19^4 \bmod 23 = 16^2 \bmod 23 = 3$
$19^6 \bmod 23 = (19^4 \times 19^2) \bmod 23$
$= (3 \times 16) \bmod 23$

$\boxed{K = 2}$

user B calculates the secret using A's public key:-

$$K = A^b \bmod p$$
$$K = 8^{15} \bmod 23$$
$8^2 \bmod 23 = 18$
$8^4 \bmod 23 = 2$
$8^8 \bmod 23 = 4$
$8^{15} \bmod 23 = (8^8 \times 8^4 \times 8^2 \times 8) \bmod 23$
$= (4 \times 2 \times 18 \times 8) \bmod 23$

$\boxed{K = 2}$

Interpretation of Results:-

Both users Independently calculate the same shared secret key, $K = 2$ without transmitting the secret itself. However, basic Diffie-Hellman does not authentica the communicating parties An attacker can Intercept and replace public keys unless authentication is used.

Conclusion:-

Digital signatures, certificates and authenticated key exchange help prevent man-in-the middle attacks. The given values demonstrate the mathematical process but are too small for real-world security.